

ME 210: Intro to Mechatronics

Pre Lab 2

Koichi Kimoto

Question 0.1

Decide which Arduino ports and pins you will use to control the motors (all parts of the lab).

Part 1

- **5V pin** to potentiometer
- **GND pin** to potentiometer
- Potentiometer wiper to **A1** (to be fed analog voltage input)
- **D7 pin** to output our DIY PWM onto motor

Part 2

- Same pins to potentiometer
- **D6 pin** (PWM pin) as outputting square wave with `analogWrite()` frequency connected to the 5V ENA pin on the L298.
- **D5 pin** going to IN1.
- **D4 pin** going to IN2.
- These set the logic low and high, determining motor spin direction on the L298, and can also “brake” the motor by setting to the same voltage.
- **GND** to GND of the L298.

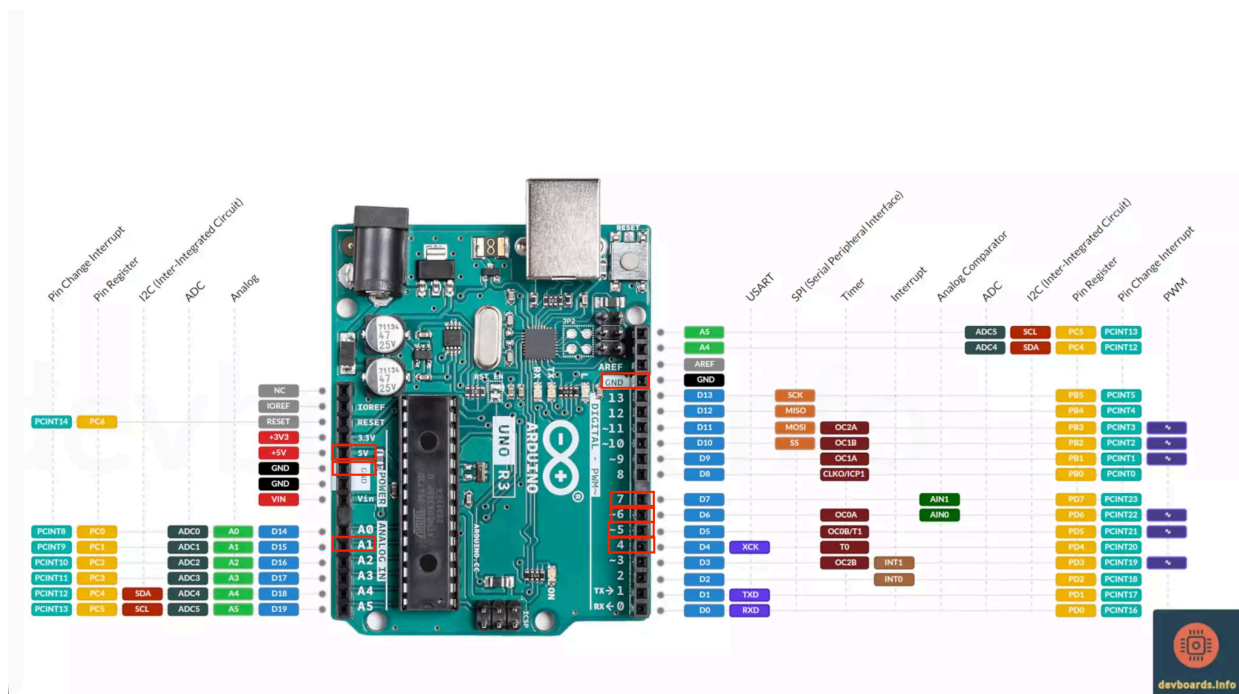


Figure 1: Selected pins (boxed in red)

Question 0.2

Decide what, if any, initialization is required for each pin.

- Part 1:
 - Initialize D7 pin as an output pin
- Part 2:
 - Initialize D4 and D5 as output pins.